

SynCocreate

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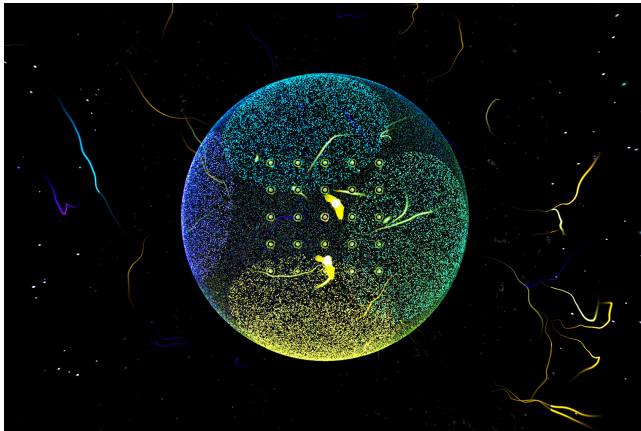


Figure 1: SynCocreate.

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1 Abstract

SynCocreate harnesses the collaborative potential of neurofeedback to offer a playful and engaging experience for cocreating artwork within a virtual 3D environment by synchronizing brainwaves between pairs of users.

The exploration of collaborative artwork and co-creation in human interactions spans from the classic exquisite corpse game to

traditional forms like visual art, musical composition, and choreography. These practices are known to foster social wellbeing such as enhancing cooperation and reinforcing bonds. Meanwhile, measurable biosignals like EEG relay internal activities and emotional states, bearing prosocial potential for interaction. However, expressing and sharing these signals, especially in remote social contexts, remains challenging.

SynCocreate addresses this challenge by unfolding within a dome that symbolizes the brain. Inside, two spectral, smoke-like sprites dance in the space, each guided by the real-time brainwave signals of a user. As the users' brainwaves synchronize, a splash of paint materializes on the dome's surface like a firework. Each user's environment incorporates visual cues — such as light, color, and pulsations synchronized with their paired partner's brainwaves — to encourage closer brainwave alignment. With each synchronization, more paint is added, gradually creating a vibrant, unique painting on the dome's interior, resembling a sky filled with fireworks, celebrating their connection. Beyond providing a mesmerizing shared artistic experience through brainwave signals, SynCocreate explores the impact of collaborative neurofeedback on co-creativity and emotional connection and investigates the role of immersive and collective visualizations of brainwaves in fostering creativity and togetherness.

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